

Governance-Gated Credit-Risk Benchmarking: Empirical-Bayes Segment Shrinkage on the UCI Default of Credit Card Clients Dataset

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github.com/asynced24/bayesian-credit-risk-gate (v1.0.0)

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Abstract

Model selection and model approval ask different questions of a credit-risk model, and they are usually answered by different artefacts at different times. This report describes a pipeline that answers both in one run. Three credit-default models are benchmarked on the UCI *Default of Credit Card Clients* dataset and each is then put through five governance gates — fairness, drift, calibration, uncertainty, and a Bayesian segment-stability check — whose thresholds are declared in configuration and validated at startup. Segment-level default rates, the unit most governance questions are posed in, are estimated with an empirical-Bayes Beta-Binomial model rather than taken raw, so that cells holding a handful of borrowers get a defensible rate and an honest interval instead of noise. On the committed offline sample all three models fail the combined verdict. The binding finding is an equalized-odds gap of 0.1617 on **sex** for logistic regression, against a fail limit of 0.10 — produced by a model that never received **sex** as a feature. Its 95% interval is [0.050, 0.274], so the point estimate clears the limit by about one standard error, and on the full 30,000-row dataset the same gap falls to 0.0245 while the fairness gate still fails, on **age_band**: the mechanism reproduces, the magnitude does not. A second finding is that governance verdicts reproduce across platforms while the benchmark ranking does not: XGBoost’s histogram tree builder moves ROC-AUC by more than the margin separating the top two models.

1 Problem statement

A credit-scoring model has to pass model-risk review before it can inform a lending decision, and review asks a different set of questions from the ones model selection asks. Selection asks which model ranks borrowers best. Review asks whether a predicted probability means what it claims, whether error rates are comparable across demographic groups, whether the input distribution has moved since training, and whether an individual adverse decision can be explained to the person who received it.

Those two sets of questions are normally answered by separate artefacts: a training script produces the discrimination metrics, and a document assembled weeks later addresses the review questions, often from summary statistics rather than from a check that runs. The separation matters because the two can disagree, and a model with the best ROC-AUC in a benchmark can still be one that should not be deployed.

This framework runs both in the same pipeline. Every threshold lives in `config.yaml`, is validated by pydantic at startup, and yields a pass/warn/fail verdict on every run; the worst gate becomes the model’s verdict. Selection and approval are separate outputs of a single execution.

A second concern is segment granularity. Governance questions are usually posed per demographic segment, and a segment default rate computed directly is dominated by sampling noise once the cell is small: in the run reported here, one segment holds six borrowers and zero defaults, and its raw default rate of 0.000 is not a usable input to any provisioning decision. Section 3 describes the empirical-Bayes model used in place of raw rates.

2 Architecture

Figure 1 shows the pipeline. The loader fetches UCI dataset 350, caches it, and can instead read a committed 5,000-row class-stratified sample so that the whole pipeline runs with no network access. The processor engineers 28 behavioural features from the six months of billing and repayment history, giving 49 predictors, and splits the data. It also separates *audit columns* from the feature matrix: **sex** and **marriage** are withheld from training entirely as prohibited bases and retained only so the fairness gate can measure against them.

Two components then feed the gates. The benchmark suite fits the three models and produces test-set predictions and metrics. Independently, the shrinkage model fits a population prior and per-segment posteriors on the training split. The five gates consume both, and each returns a pass/warn/fail verdict

with the metric and threshold that produced it. Explainability and reporting run after the gates, and every parameter and metric is logged to MLflow or, when MLflow is disabled, to a local JSON run log.

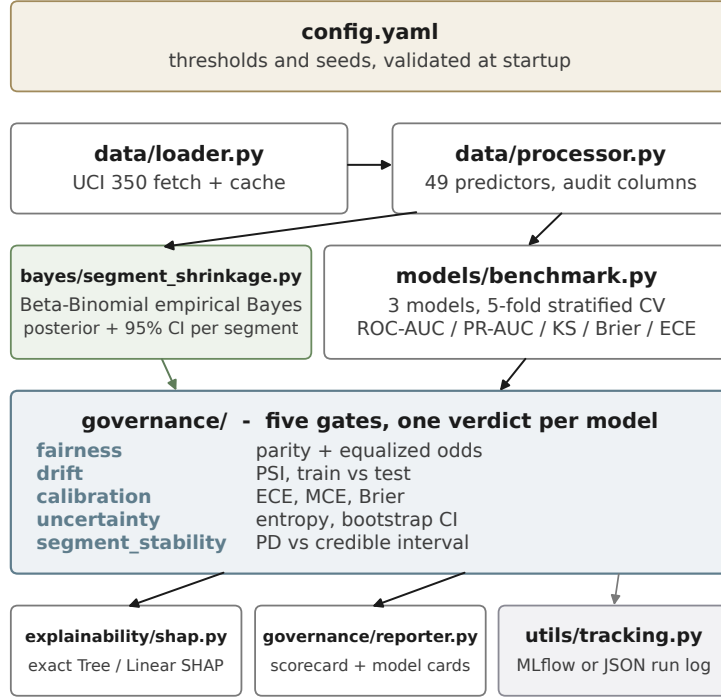


Figure 1: Pipeline stages. Thresholds, seeds and the model list are declared once in `config.yaml` and read by every stage below it. The benchmark suite and the empirical-Bayes shrinkage model are independent inputs to the gate layer: the first supplies predicted probabilities, the second the segment posteriors the stability gate compares them against.

3 Empirical-Bayes segment shrinkage

Segments are defined as the cross of education level and age band. Education has four levels once the codes the source paper leaves undocumented are folded into its “other” category, and age is binned on the edges 20, 30, 40, 50, 60, 80 declared in `config.yaml`, left-closed and right-open, giving the five bands 20–30 through 60–80. The cross is chosen because education and age are the two demographic columns the model is actually allowed to use as predictors: `sex` and `marriage` are withheld as prohibited bases, so segmenting on them would define the governance unit in terms of a column no lending decision may depend on, and segmenting on a behavioural feature would make the reference cell a function of the same history the model is scoring. Four levels times five bands is 20 possible cells, of which 19 appear here — `other_unknown` | 60–80 holds four borrowers in the full 30,000-row dataset and none of them are drawn into the committed 5,000-row sample, so the cell is absent rather than dropped.

Rather than take each cell’s observed default rate, the framework treats the counts hierarchically: each segment has its own latent default rate, and those rates are themselves draws from one population-level distribution. Because the Beta is conjugate to the Binomial, every segment’s posterior is closed form, and its mean is automatically a precision-weighted blend of the segment’s own rate and the population rate. The weight is not a tuned constant — it falls out of the posterior algebra, so a thin cell shrinks hard toward the population rate and a well-populated cell barely moves. For segment i with n_i borrowers and k_i defaults:

$$\theta_i \sim \text{Beta}(\alpha, \beta), \quad k_i \sim \text{Binomial}(n_i, \theta_i), \quad (1)$$

$$\theta_i \mid k_i, n_i \sim \text{Beta}(\alpha + k_i, \beta + n_i - k_i), \quad (2)$$

$$\mathbb{E}[\theta_i \mid k_i, n_i] = w_i \frac{k_i}{n_i} + (1 - w_i) \frac{\alpha}{\alpha + \beta}, \quad w_i = \frac{n_i}{n_i + \alpha + \beta}. \quad (3)$$

The 95% credible interval is read off the same posterior as the 2.5th and 97.5th percentiles of $\text{Beta}(\alpha + k_i, \beta + n_i - k_i)$.

The prior (α, β) is estimated by method of moments across segments, with one correction that matters. The observed spread of the empirical rates $p_i = k_i/n_i$ overstates the spread of the underlying θ_i , because each p_i carries its own binomial sampling noise:

$$\text{Var}(p_i) = \text{Var}(\theta_i) + \mathbb{E} \left[\frac{\theta_i(1 - \theta_i)}{n_i} \right]. \quad (4)$$

Subtracting the sampling component gives $\text{Var}(\theta_i) = \text{Var}(p_i) - \overline{\mu(1 - \mu)/n_i}$, floored at a small positive constant, and the Beta concentration then follows from the identity $\text{Var}(\theta_i) = \mu(1 - \mu)/(\alpha + \beta + 1)$. Using $\text{Var}(p_i)$ directly instead yields a prior that is too diffuse and therefore under-shrinks. Only segments with $n_i \geq 30$ contribute to the prior fit; every segment receives a posterior regardless.

On the run reported here, 12 of the 19 segments cleared the prior-fitting floor. The pooled rate is $\mu = 0.2230$, the observed variance of segment rates is 1.4581×10^{-3} and the estimated binomial sampling variance is 1.0243×10^{-3} , leaving $\text{Var}(\theta_i) = 4.3375 \times 10^{-4}$. That gives a concentration of $\mu(1 - \mu)/\text{Var}(\theta_i) - 1 = 398.49$ and a prior of Beta(88.87, 309.62) — in other words, a prior worth about 398 equivalent borrowers.

$\text{Var}(p_i)$ is taken unweighted across those 12 segments, which is the consistent reading of the model: each segment contributes one draw of θ_i , so weighting the spread by n_i would estimate the dispersion of θ across borrowers rather than across segments, and would let the two or three largest cells set the prior for all of them. The choice is not free numerically — size-weighting the same 12 segments, sampling term included, gives a concentration of 352.70 rather than 398.49, a prior worth about 11% fewer equivalent borrowers — but it moves no posterior far enough to change a gate verdict in this run.

Two limitations of the construction are worth stating rather than leaving to be noticed. First, (α, β) is estimated from 12 segments and then treated as known when the posteriors and their credible intervals are formed. The intervals reported here are therefore conditional on $(\hat{\alpha}, \hat{\beta})$ and understate the true uncertainty by whatever is attributable to having estimated the hyperparameters from a small number of cells — the standard critique of empirical Bayes [3]. A fully Bayesian treatment with a hyperprior, or a parametric bootstrap over the prior fit, would widen them; on 12 segments the correction would not be negligible. Second, the model assumes the segment rates are exchangeable draws from one population distribution. That is an assumption about education $\times$ age-band cells, not a property of them that this run tests, and no covariate structure enters the prior: a 60-something with a graduate degree borrows the same prior as a 20-something who did not finish high school. A covariate-adjusted small-area model — Fay-Herriot, say, letting the prior mean depend on education and age directly — is the natural extension, and it would matter most for exactly the thin cells shrinkage exists to rescue.

Figure 2 shows the effect. The seven segments left of the boundary sit below the prior-fitting floor, and their raw rates scatter from 0.000 to 0.333 while their posteriors all sit within a few points of the population rate with wide intervals. The thinnest, `other_unknown | 50-60`, has $n = 6$ and $k = 0$: its posterior is Beta(88.87, 315.62), mean 0.220, interval [0.181, 0.261], and by Equation 3 the weight on its own data is $6/(6 + 398.49) = 0.01$. To the right, posteriors track their raw rates closely and the intervals tighten. Across all 19 segments the median interval width is 0.0732 and the median weight on own data is 0.212.

The segment-stability gate then compares each model’s mean predicted probability of default per segment against these intervals, and flags any segment where the model prices the cell differently from its noise-corrected historical experience.

4 Benchmark results

Table 1 and Figure 3 report a run of `python main.py -sample` at seed 42 on the committed sample (hash `b1ba275532b83033`, base rate 0.2212), 3,750 train and 1,250 test rows, 49 predictors. The split is stratified on the target, so both halves carry the base rate to three decimal places (0.221 train, 0.221 test) and the cross-validation folds within the training split are stratified as well.

Class weights are left at their natural values throughout, because rebalancing buys nothing measurable here and costs the calibration that the ECE and Brier gates exist to measure. Refitting with `class_weight="balanced"` moves test ROC-AUC by +0.0016 for logistic regression and −0.0033 for LightGBM: not only small against the fold-to-fold standard deviations below, but not even signed consistently between the two models, which is the general case rather than a surprise. Calibration is the half of that trade that is one-directional. Logistic regression’s ECE goes from 0.0333 to 0.2116 and LightGBM’s from 0.0411 to 0.1303, with mean predicted PD rising from 0.217 and 0.206 to 0.432 and 0.342 against an observed 0.221 — and a probability of default that no longer means “probability of default” is not usable for provisioning.

The three models span 0.0026 of ROC-AUC against fold-to-fold standard deviations of 0.0154 to 0.0224. Reading that as a ranking would be overreading it, which is the practical argument for reporting the gate

Empirical-Bayes shrinkage over 19 education × age-band segments
prior Beta(88.87, 309.62), worth 398 equivalent borrowers

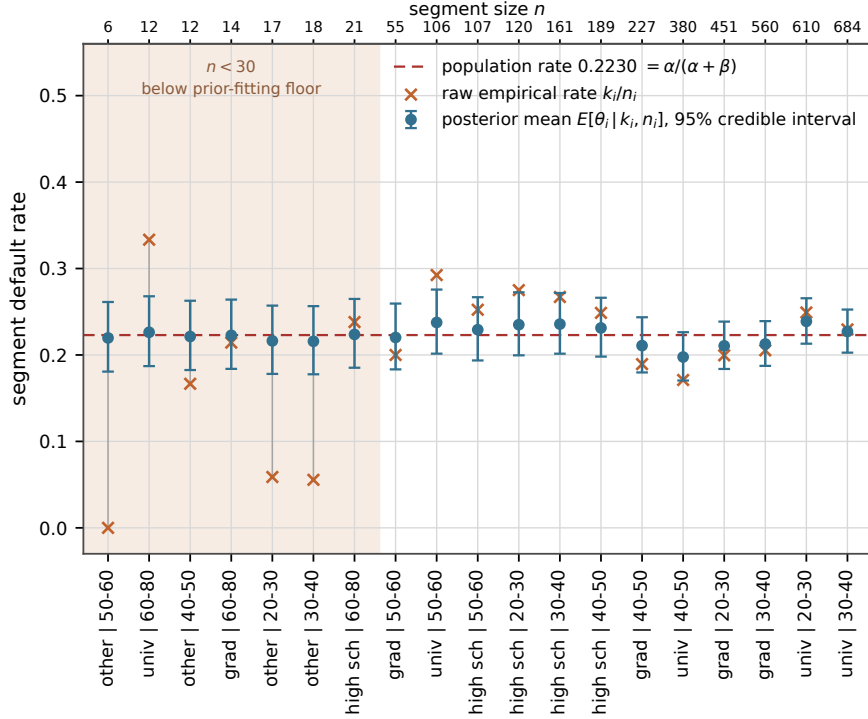


Figure 2: Raw versus shrunk segment default rates, 19 education × age-band segments ordered by segment size (sizes on the top axis). Crosses are the raw empirical rates k_i/n_i ; circles are posterior means with 95% credible intervals; the grey stems show how far shrinkage moved each estimate. Thin segments (shaded, $n < 30$) collapse toward the population rate with wide intervals; well-populated segments barely move. Produced by fitting the model on the training split of the committed sample.

Table 1: Benchmark and gate verdicts, committed offline sample, seed 42. Discrimination metrics are test-set; CV columns are 5-fold stratified cross-validation on the training split. This run is Windows / Python 3.13.7; `logistic_regression` and `lightgbm` are bit-identical on Linux, `xgboost` is not (Section 6).

Model	ROC-AUC	PR-AUC	KS	Brier	Log-loss	ECE	CV mean	CV s.d.	Verdict
<code>lightgbm</code>	0.7634	0.5355	0.4220	0.1374	0.4469	0.0411	0.7671	0.0179	FAIL
<code>xgboost</code>	0.7625	0.5318	0.4215	0.1379	0.4474	0.0438	0.7653	0.0154	FAIL
<code>logistic_regression</code>	0.7607	0.5042	0.4155	0.1403	0.4459	0.0333	0.7511	0.0224	FAIL

matrix alongside the leaderboard rather than beneath it. Gradient boosting does earn a real margin on PR-AUC — 0.5355 against 0.5042 — so the tree models are better at concentrating the minority class, even though their ordering between themselves is not resolvable at this sample size.

5 Governance findings

Table 2 gives the gate matrix. The overall verdict is FAIL for every model, which is the intended outcome rather than a broken build: a governance harness tuned so that everything passes is not measuring anything.

Table 2: Gate verdicts. The worst gate is the model’s verdict.

Gate	<code>lightgbm</code>	<code>xgboost</code>	<code>logistic_regression</code>
fairness	FAIL	FAIL	FAIL
drift	PASS	PASS	PASS
calibration	WARN	WARN	WARN
uncertainty	PASS	PASS	PASS
segment_stability	FAIL	FAIL	WARN

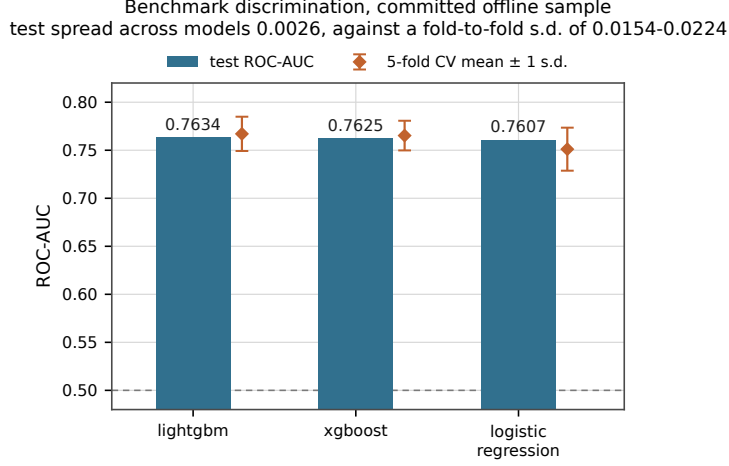


Figure 3: Test ROC-AUC per model with the cross-validation mean and standard deviation alongside; the dashed line marks chance at 0.50. The spread between models is roughly an order of magnitude smaller than the fold-to-fold spread within each model.

5.1 Fairness: a gap on an attribute the model never saw

The sharpest result is on **sex** under logistic regression. **sex** is withheld from the feature matrix entirely, along with **marriage**, and kept only as an audit column. The model nonetheless produces an equalized-odds gap [4] — here the larger of the max-min true-positive-rate spread and the max-min false-positive-rate spread across groups — of **0.1617**, past the 0.10 fail limit, while passing demographic parity at 0.0459. Table 3 shows why: the gap is almost entirely in the true-positive rate, where the false-positive-rate spread is only 0.0084.

Table 3: Fairness detail for **sex**, logistic regression, decision threshold 0.5. Demographic parity gap 0.0459 (pass); equalized-odds gap **0.1617** (fail, limit 0.10), 95% interval [0.050, 0.274]. n_{def} is the number of actual defaulters in the group, which is what the TPR is estimated on.

sex	n	n_{def}	Predicted default rate	Observed default rate	TPR	FPR
male (1)	495	113	0.1333	0.2283	0.4071	0.0524
female (2)	755	163	0.0874	0.2159	0.2454	0.0439

That gap is a difference of two rates estimated on thin cells, and a report whose argument is that raw rates from small cells need intervals owes its own headline the same treatment. The TPR difference rests on 113 male and 163 female defaulters in the test split. The Wald standard error for the difference of two independent proportions is

$$\text{s.e.}(\Delta\text{TPR}) = \sqrt{\frac{\hat{p}_m(1 - \hat{p}_m)}{n_m} + \frac{\hat{p}_f(1 - \hat{p}_f)}{n_f}} = 0.0572, \quad (5)$$

giving a 95% interval of [0.050, 0.274] around the 0.1617 point estimate. A 20,000-draw nonparametric bootstrap over the same rows, resampling within each (sex, outcome) cell, agrees closely: standard error 0.0570, percentile interval [0.050, 0.273]. The point estimate therefore sits 1.08 standard errors above the 0.10 fail limit, and the interval comfortably contains values below it.

Two readings follow, and both are worth keeping. The gate outcome is real: the threshold is defined on the point estimate, the point estimate breaches it, and a gate that waited for statistical significance before flagging a candidate disparity would be the wrong instrument for the job — the cost of a false-positive FAIL is a review, the cost of a false-negative PASS is a deployed model. But the evidence for a gap larger than 0.10 *specifically* is weak, and the finding should not be quoted as though 0.1617 were pinned down. What is pinned down is the direction and the mechanism, not the magnitude.

The gap also depends on where the decision threshold is put, which matters here because 0.5 is a long way into the tail of a book with a 0.221 base rate: only 10.6% of the test set is classified as a default at that threshold. Table 4 recomputes the same gap at 0.30 and at the base-rate-matched 0.3153, the threshold that classifies exactly as many borrowers positive as actually default. The direction survives at every threshold; the crossing of the 0.10 limit does not. 0.5 is the configured operating point (`models.decision_threshold`) and is where the gate runs, but a reviewer reading Table 3 should know that the FAIL verdict is not robust to a defensible alternative choice of threshold.

Table 4: Threshold sensitivity of the `sex` true-positive-rate gap, logistic regression. Positive rate is the share of the test set classified as a default. The base-rate-matched threshold is the $(1 - 0.2208)$ quantile of the predicted probabilities. Standard errors are from Equation 5.

Threshold	Positive rate	TPR male	TPR female	Δ TPR	s.e.	Gate
0.30	0.2296	0.6018	0.5092	0.0926	0.0604	WARN
0.3153 (base-rate)	0.2208	0.6018	0.5031	0.0987	0.0605	WARN
0.50 (configured)	0.1056	0.4071	0.2454	0.1617	0.0572	FAIL

Two further things follow from Table 3 itself. First, a parity-only fairness check would have signed this model off: predicted default rates are close across the two groups, and it is recall on borrowers who actually defaulted that is not — a defaulting male applicant is identified 41% of the time against 25% for a defaulting female applicant. Second, and more to the point, the model produced that gap without ever reading the column, which is consistent with correlated behavioural features carrying the signal back in.

That last clause is a hypothesis, so it is worth at least partially checking rather than asserting. Training a model to recover `sex` from the same 49 retained predictors gives a test ROC-AUC of 0.588 for logistic regression and 0.607 for LightGBM, against a 60.4% female base rate. `sex` is therefore weakly but not negligibly recoverable from the features that survive exclusion: enough for the repayment history to be a plausible route for the gap, not enough to establish it as the route. Identifying the actual channel would need a feature ablation this pipeline does not run, and the claim is left at that. What the result does establish is the case that dropping a protected attribute does not cover, and the reason the gate audits withheld attributes rather than treating exclusion as sufficient. Logistic regression is also one of the two models that reproduce bit-identically across platforms, so the figure is not an artefact of the machine it was computed on.

The framework audits `age_band` as well, and the contrast is instructive rather than equivalent. `age` is used directly as a model feature — 12th of 49 by SHAP importance on LightGBM (Section 7) — and its equalized-odds gap of 0.1600 on LightGBM therefore reflects the model using an input it was handed. That raises the question of whether the use is justified; it is not evidence of a hidden proxy in the way the `sex` result is. The 60–80 band is excluded from the gate entirely: 10 test rows is below the configured `min_group_size` of 50, and reporting a true-positive rate on that is worse than reporting nothing.

5.2 Calibration, drift and uncertainty

Calibration warns on all three models. LightGBM’s expected calibration error [6] — the bin-count-weighted mean absolute difference between predicted probability and observed frequency over 15 equal-width reliability bins — is 0.0411 against a warn limit of 0.03, with maximum calibration error 0.2266 over the 14 bins that are populated and a mean predicted PD of 0.2060 against an observed rate of 0.2208: a slight systematic under-prediction of 0.0148. Drift passes cleanly, as it should on a random split: maximum feature PSI is 0.0156 across all 49 features, on `payment_ratio_mean`, with no feature over the 0.10 warn line. Uncertainty passes with 17.8% of the test set above the 0.60-nat predictive-entropy threshold, against a warn line at 35% and a fail line at 55% — so the gate is not close to firing, and the figure is reported because the share of the book the model refuses to call is worth knowing even when it is acceptable. The bootstrap 95% interval for LightGBM’s ROC-AUC is $[0.7022, 0.8097]$ — a width of 0.107, which is worth holding alongside the 0.0026 spread that separates the three models.

5.3 Segment stability

The stability gate fails LightGBM: 4 of 11 gated segments price outside their credible interval, a ratio of 0.364 against a fail limit of 0.30. The clearest breach is `graduate_school` | 30–40, where the model’s mean predicted PD is 0.163 against a shrunk posterior mean of 0.213 with interval $[0.187, 0.239]$, built from 560 reference borrowers. `university` | 50–60 breaches in the same direction, 0.187 against $[0.202, 0.276]$. Both are under-predictions: the model prices default risk for these cells below their noise-corrected historical experience. That is a provisioning concern, and no aggregate metric in Table 1 surfaces it. Logistic regression warns rather than fails here, at 0.273.

5.4 Robustness on the full 30,000 rows

Everything above is computed on the committed 5,000-row stratified sample, which exists so that the pipeline, its tests and CI run without a network call. Rerunning the identical configuration against the full 30,000-row dataset is the obvious robustness check, and it changes what the headline can honestly be said to be.

Most of the picture holds or improves. The combined verdict is still FAIL for all three models, and still on the fairness gate, for the same reason in kind: an equalized-odds breach on a demographic attribute. Discrimination rises with the extra data, to ROC-AUC 0.7781 for both LightGBM and XGBoost and 0.7573

for logistic regression, on fold-to-fold standard deviations of 0.0069–0.0086 rather than the 0.0154–0.0224 of Table 1. Calibration moves from WARN to PASS on all three (ECE 0.0164, 0.0156, 0.0193), which is what more data should do to a calibration warning. Drift and uncertainty pass as before. Segment stability passes for the tree models and warns for logistic regression, and all 20 education $\times$ age-band cells are populated, so nothing is dropped.

The `sex` result does not survive at its sample magnitude. On the full data the logistic-regression equalized-odds gap on `sex` falls to 0.0245, with a Wald 95% interval of $[-0.021, 0.070]$ on 702 male and 957 female test-set defaulters — the same sign, roughly a sixth of the size, and a pass rather than a fail. The two intervals overlap, so the estimates are not in formal contradiction; 0.1617 with a standard error of 0.0572 is simply what an estimate near a threshold does on 276 defaulters. What binds instead on the full data is `age_band`, at 0.1121 to 0.1417 across the three models: a failing gap on the attribute the model is handed, not on the one withheld from it.

The reading that survives is that the mechanism is worth taking seriously and the specific number is not. A withheld attribute can produce a measurable error-rate gap; the gate finds it without being told to look for a proxy; and the same audit run at six times the sample size still fails, on a different attribute. That the `sex` effect is small enough on 30,000 rows to pass a 0.10 limit is a more useful finding than the sample figure was, and it is the argument this report makes for shrinking segment rates, turned on its own headline. Regenerating Tables 1–4 against the full dataset, rather than reporting it in prose here, is left as the next revision; the figures and the worked examples throughout remain those of the offline sample.

6 Cross-platform reproducibility

On the same seed and the same sample hash, across Windows and Linux:

- `logistic_regression` and `lightgbm` reproduce **bit-identically** — every metric, every gate verdict, every credible interval.
- The Bayesian layer reproduces bit-identically everywhere as well. That is an observation about these two platforms, not a guarantee: the layer runs no threaded reduction of its own, which removes the specific cause behind the XGBoost result below, but cross-platform bit-identity ultimately turns on the BLAS build and the floating-point behaviour underneath it. Being numpy/scipy only makes the outcome likely, not certain.
- `xgboost` does **not**. Its histogram tree builder reduces floating-point sums in a thread- and platform-dependent order, so the same seed gives ROC-AUC 0.7672 on Linux and 0.7625 on Windows, with ECE 0.0320 against 0.0438.

That 0.0047 spread is larger than the 0.0038 margin between the top two models on Linux, so *which model wins depends on the machine it is run on*: XGBoost on Linux, LightGBM on Windows. Table 1 reports the Windows run for that reason, and the CI table in the repository README reports the Linux run.

Every gate verdict in Table 2 is identical on both platforms, which is the part worth keeping: the approval decision is stable even where the leaderboard is not. A governance verdict that flipped with the host would be useless, whereas a benchmark ranking that flips is merely unreliable. Anyone needing a stable winner as well as a stable verdict should pin the platform or set `models.enabled` to a single model.

7 Explainability

SHAP runs on whichever model the selection metric picks, which on this run is LightGBM. Explainer choice is by model family — exact tree SHAP for the boosters, exact linear SHAP for the scaled logistic pipeline — so neither path uses sampling approximation [5]. Both output on the log-odds scale, where contributions are additive: $\sigma(\text{base} + \sum \text{SHAP})$ recovers the predicted probability exactly. A mean absolute SHAP value is therefore movement in log-odds, not a change in probability of default.

Figure 4 shows the result. Most recent repayment status (`pay_status_1`, 0.359) dominates at more than twice the next feature, followed by `max_delinquency` (0.152), `limit_bal` (0.135) and `delinquent_months` (0.122). Delinquency history and credit limit carrying the model is consistent with the source paper [1]. The pipeline also writes per-feature contribution breakdowns for three individual borrowers — the highest-risk, the lowest-risk, and the most uncertain in the sample — as JSON, in the form an adverse-action review needs.

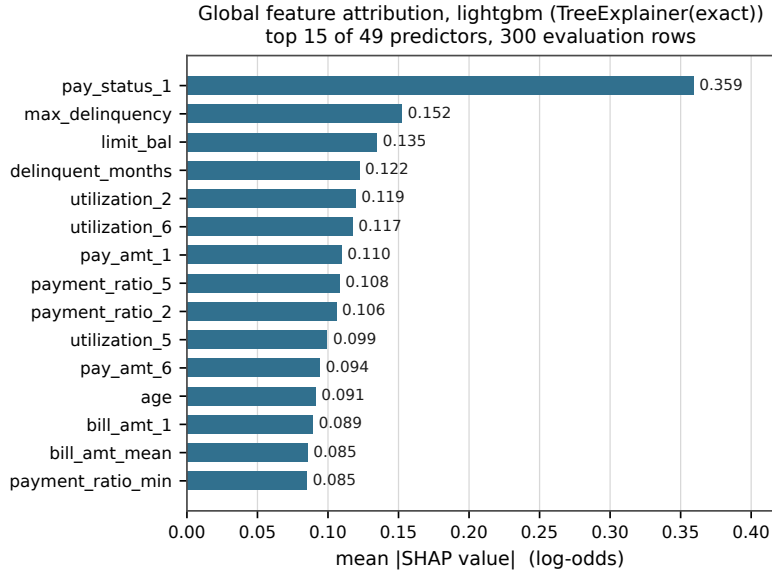


Figure 4: Global feature attribution for LightGBM, exact `TreeExplainer`, top 15 of 49 predictors over 300 evaluation rows against 100 background rows.

8 Dataset, tech stack, and reproducing this report

The data is UCI Machine Learning Repository dataset 350, *Default of Credit Card Clients* [1, 2]: 30,000 Taiwanese credit-card customers, 23 features covering six months of billing and repayment history plus limited demographics, and a binary target for default on the next payment, at a base default rate of 22.1%. It is distributed under CC BY 4.0; the repository commits a 5,000-row class-stratified draw (base rate 22.12%) under the same licence, so the pipeline, its tests, and the CI smoke job run with no network call.

Models are scikit-learn `LogisticRegression`, `xgboost` and `lightgbm`; the Bayesian layer is numpy and scipy only; configuration is pydantic v2 with `extra="forbid"`; explainability is `shap`; tracking is MLflow with a local JSON fallback. CI runs ruff, pytest on Python 3.12 and 3.13, and an end-to-end smoke job that asserts both scorecards and all three model cards were written, that three models were benchmarked with five gates each, that every ROC-AUC beat chance, and that the shrinkage prior was fitted.

Everything in this report regenerates from the tagged release:

```
git clone https://github.com/asynced24/bayesian-credit-risk-gate.git
cd bayesian-credit-risk-gate && git checkout v1.0.0
pip install -r requirements.txt
python main.py -sample
python docs/figures/generate_figures.py
python main.py    (full 30,000 rows, Section 5.4; needs network once)
```

Tables 1–4 are taken from the scorecard JSON that the `-sample` run writes into `reports/`. The four figures come from `generate_figures.py`, and Figure 2 re-fits the shrinkage model directly rather than reading a cached artefact. The final command fetches and caches the full dataset and produces the run reported in Section 5.4; every subsequent invocation reads the cache. Python 3.12 or newer is required, since the pinned numpy, scipy, xgboost and shap releases all declare `Requires-Python >= 3.12`.

References

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